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EE 471C - HW - 2

1. $h(t) = \delta(t-t_0) + \alpha\delta(t-\tau_1)$

$x_r(t) = \sum_n x[n]\,\text{sinc}\left(\frac{t-nT}{T}\right)$ generalized ideal D/C converter. (interpolator).

Using equations from slides ~~(and $W=1$ assumed)~~.

$h[n] = \int \text{sinc}(\tau W)\left[\delta(nT-\tau-t_0)+\alpha\delta(nT-\tau-\tau_1)\right]d\tau$

General Ans. $\longrightarrow = \text{sinc}((nT-t_0)W) + \alpha\,\text{sinc}((nT-\tau_1)W)$

Assuming Nyquist ($TW=1$) sampling:

$W = \frac{1}{T}$, so $(nT-\tau_0)W = nTW - t_0W$

$n(1) - t_0W.$

$n - \frac{\tau_0}{T}$

$h[n] = \text{sinc}\left(n - \frac{\tau_0}{T}\right) + \alpha\,\text{sinc}\left(n - \frac{\tau_1}{T}\right)$

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e^{-3t} $u = -3t$
$du = -3\,dt$
$-\frac{1}{3}-$

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PURPOSE

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NOTES

2) $R_{xx}(\tau) = e^{-2\beta|\tau|}$

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau)\, e^{-j\omega\tau}\, d\tau$$

$$S_{xx}(\omega) = \int_{-\infty}^{0} e^{2\beta\tau} e^{-j\omega\tau}\, d\tau + \int_{0}^{\infty} e^{-2\beta\tau} e^{-j\omega\tau}\, d\tau$$

$$\left[\frac{e^{(2\beta - j\omega)\tau}}{(2\beta - j\omega)}\right]\Bigg|_{-\infty}^{0} \qquad \left\{\frac{e^{(-2\beta - j\omega)\tau}}{(-2\beta - j\omega)}\right\}$$

$e^{(2\beta - j\omega)(-\infty)} \to 0$

$e^{(2\beta - j\omega)(\tau = 0)} \to 1$

$$\frac{1}{2\beta - j\omega} + \frac{1}{2\beta + j\omega} = \frac{(2\beta + j\omega) + (2\beta - j\omega)}{(2\beta - j\omega)(2\beta + j\omega)}$$

$$= \frac{4\beta}{4\beta^2 - (j\omega)^2}$$

$$S_{xx}(\omega) = \boxed{= \frac{4\beta}{4\beta^2 + \omega^2}}$$

SUMMARY

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B) $\text{delay} = \frac{100m}{3E8\ m/s} = 3.\overline{33}E-7$ or 333.3 ns

A) $h_c(t) = \alpha\,\delta(t-\tau)$ attenuation and delay are known, so

$\boxed{h_c(t) = 0.1\,\delta(t - 333.3\text{ ns})}$ [given t in ns.

B) Passband is also just the LTI model - $h_c(t)$ has NO frequency-distortion.

$\boxed{h_p(t) = 0.1\,\delta(t - 333.3_{ns})}$

C) $y_p(t) = \alpha\,\mathrm{Re}\{x(t)e^{j2\pi f_c(t-\tau)}\}$

$= \mathrm{Re}\{\alpha\,x(t-\tau)e^{-j2\pi f_c\tau}\,e^{j2\pi f_c t}\}$ ← distribute

for impulse resp. case,

$h(t) = \alpha e^{-j2\pi f_c t}\,\delta(t-\tau)$.

$f_c\cdot\tau = (2E9)(3.33E-7) = 666.\overline{66}$ cycles

$\theta = -2\pi(0.667) = 120°$

$\boxed{h(t) = \overset{\alpha}{0.1}\,e^{+j(120°)}\,\delta(t - 333.3\text{ ns})}$

or $0.1e^{-j2\pi(666.\overline{6})}\,\delta(t - 3.33E-7)$ all in seconds

D) $T = \frac{1}{5E6} = 200\text{ ns}$ (Bandwidth given as 5 MHz)

Discrete sequence is sinc-interpolated, so,

$h[n] = \alpha e^{-j2\pi f_c\tau}\,\mathrm{sinc}\left(n - \frac{\tau}{T}\right)$ $\quad \frac{\tau}{T}$ here $= \frac{333.3}{200}$

$= 1.66$ (non integer).

SUMMARY

$h[n] = 0.1e^{j120°}\,\mathrm{sinc}(n - 1.667)$

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4) $\mu = \frac{1}{4}\left[1 + \frac{1-j}{\sqrt{2}} - 1 + \frac{1-j}{\sqrt{2}}\right]$

$= \frac{1}{4}\left[\frac{-2j}{\sqrt{2}}\right] = \boxed{\frac{-j}{2\sqrt{2}}} \neq 0.$

Subtract mean (μ) from each symbol

a) $j - \frac{-j}{2\sqrt{2}} = \boxed{\frac{j(2\sqrt{2}+1)}{2\sqrt{2}}}$

b) ~~$j - \left(\frac{1-j}{\sqrt{2}}\right) = \frac{j\sqrt{2}}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$~~

b) $\frac{2(1-j)}{2(\sqrt{2})} + \frac{j}{2\sqrt{2}} = \frac{2(1-j)+j}{2\sqrt{2}} = \boxed{\frac{2-j}{2\sqrt{2}}}$

c) $-j + \frac{j}{2\sqrt{2}} = \boxed{\frac{-j(2\sqrt{2}-1)}{2\sqrt{2}}}$ output.

d) ~~complex conjugate of (b) by pattern recall~~ $\Rightarrow$ ~~$\frac{2-j}{2\sqrt{2}}$~~

$\frac{2(1-j)}{2(\sqrt{2})} + \frac{j}{2\sqrt{2}} = \frac{-2+2j+j}{2\sqrt{2}} = \boxed{\frac{-2-j}{2\sqrt{2}}}$

~~$\Rightarrow |a|^2, |b|^2, |c|^2 + |d|^2 =$~~

$|a|^2 + |b|^2 + |c|^2 + |d|^2 = 3.5$

$E_{Avg} = \frac{3.5}{4} = \frac{7}{8}$. To normalize and rationalize, use $\frac{\sqrt{8}}{\sqrt{7}} = \frac{2\sqrt{2}}{\sqrt{7}}$ scaling.

multiply each by $\frac{2\sqrt{2}}{7}$.

$$C_{norm} = \left\{ \frac{(1+2\sqrt{2})j}{\sqrt{7}}, \frac{2-j}{\sqrt{7}}, \frac{-j(2\sqrt{2}-1)}{\sqrt{7}}, \frac{-2-j}{\sqrt{7}} \right\}$$